ME 780: DESIGN OF A MECHATRONICS SYSTEM

DA3: RESPONSE PAPER *(Q1 & Q2 TO BE COMPLETED **INDIVIDUALLY**)*

Name: Philopatear Salib
Group Number: 8

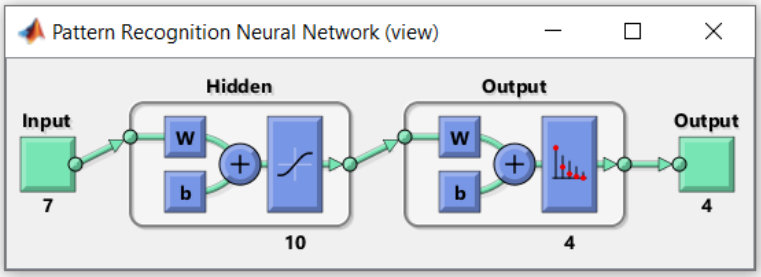
Q1. Neural Network Implementation for each group (model can be completed in groups, Q1A, B, C, D is individual)

Examine network parameters

```
view(net)
net.layers{1}
```

```
ans =
    Neural Network Layer

            name: 'Hidden'
      dimensions: 10
     distanceFcn: (none)
   distanceParam: (none)
       distances: []
         initFcn: 'initnw'
    netInputFcn: 'netsum'
  netInputParam: (none)
       positions: []
           range: [10x2 double]
            size: 10
    topologyFcn: (none)
    transferFcn: 'tansig'
  transferParam: (none)
        userdata: (your custom info)
```



DA3 Group8.zip

Qualitatively/Quantitatively describe How each part of the proposed model relates to:

A) Input Layer

The data collected in the two files: trainData and testData are the input layer.

B) Loadcase Classification

4 labels were included in our loadcase classification for the training data: 45 kg sitting, 72 kg sitting, 14 kg leaning and 19 kg leaning, 2 labels were included in our loadcase classification for the test data 58.5 kg sitting and 72 kg sitting and 19 kg leaning.

C) Life Prediction

Instead of it, using the fatigue life equation we did it analytically.

D) Output Layer

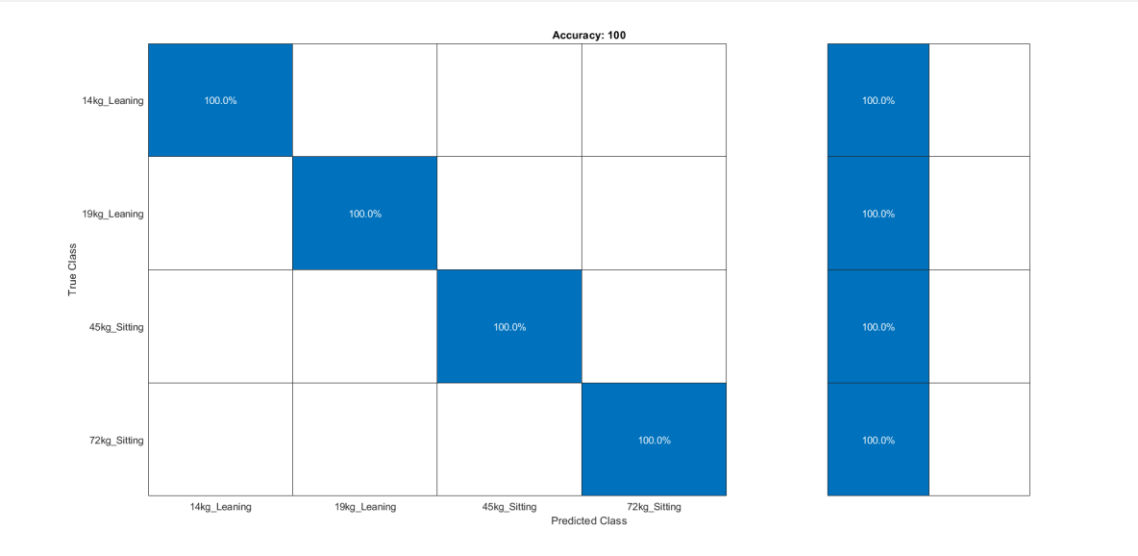
The output layer is the loadcase classification layer, outputs are probabilities for each classification.

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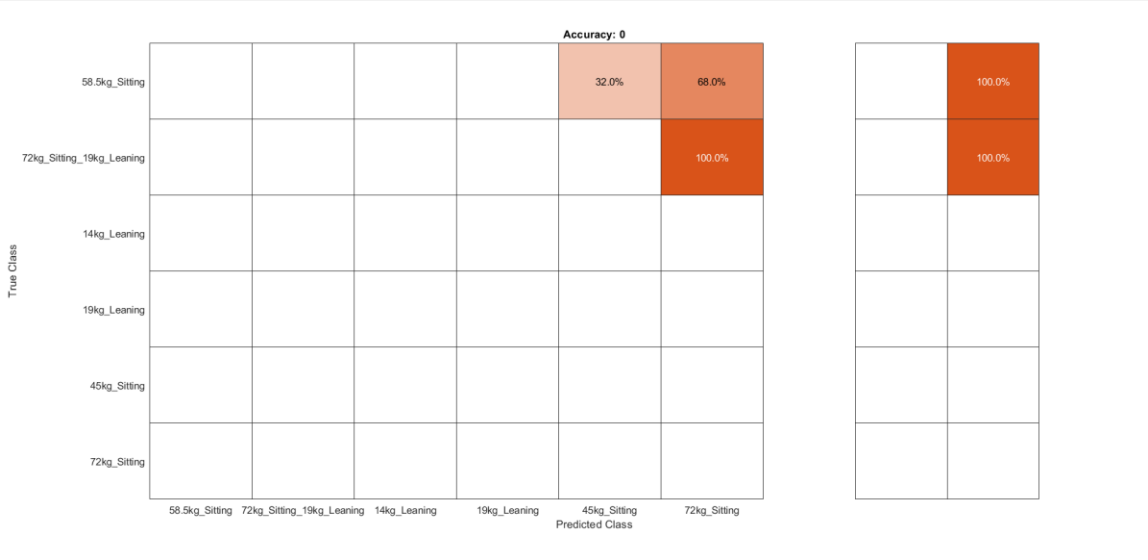
Q2. Train your ANN model using the provided training data on learn. Once your model is trained to your level chosen level of accuracy, please answer the following questions.

A) *Using the provided "ANN **Training** Data", use your model to provide a plot similar to below to show the accuracy of your loadcase classification. Comment on any patterns, or interesting trends in the result.*



A 100% accuracy was easily achieved because of the high discreteness of the training data.

B) *Using the provided "ANN **Test** Data", use your model to provide a plot similar to below to show the accuracy of your loadcase classification. Comment on any patterns, or interesting trends in the result.*



Since we did not train the model on the 58.5 kg test case, so it suggested that it may be on the 45 kg side with confidence of 32% and may be on the 72 kg side with confidence of 68%, as my understanding it should be in the middle between them 50% confidence for both sides.

Also, we did not train it for the 72 kg sitting + 19 kg leaning test case so its best match was 72 kg sitting and it suggested it correct by confidence level of 100%.

C) *Comment on any differences between the accuracy achieved when doing loadcase classification using the training data vs. the test data.*

As I mentioned before that A 100% accuracy was easily achieved because of the high discreteness of the training data.

But since we never introduced the test data to the model its accuracy turned out to be 0% but it suggested the nearest possible classification of the 4, we trained the model on.

D) *Using the provided "fatigue data" (i.e. S-N curves, stress in critical area vs. displacement), predict the amount of cycles to failure for each of the following loadcases. Comment on how each of these compares to the fatigue curve for a laboratory "material" test for the Bamboo chair material.*
   a. *45 kg Child sitting: 9235_cycles.*
   b. *72 kg adult sitting:211_cycles.*
   c. *14 kg force leaning back: 3,696,930_cycles.*
   d. *19 kg force leaning back: 282,006_cycles.*
   e. *Comments:*
      I think the variation in the experimental testing cases is due to the organic structure of the bamboo and the variation from sample to sample in all the other natural phenomena like the fiber density or the orientational changes. one of those phenomena is the fiber geometry which will impact the fatigue life and will make a high variation to the experimental fatigue.

E) *Provide justification to your input/outputs & weighting that you have selected for your neural network. Provide evidence (where applicable) to support your reasoning.*
   a. *Inputs*
      For the input data we used it all as after discussion we decide not to exclude any of them as all seems relatable, we could differentiate between sitting and leaning by using the XY coordinates and for the different weight we used the load magnitude.
   b. *Outputs*
      For the output it was the load case classification since it is its main role, and for the life cycle prediction we saw that the easiest way to do it is by analytical calculations as we do not have enough data to train our model for getting the stress and life cycle from the data.
   c. *Weighting*

      In order to improve our model, we decided that we may make two classifications to help us, specially with the test case of 72 kg sitting + 19 kg leaning, the first classification will be for the sitting load magnitude and will put more weight on the load magnitude data, and the second classification will be for the leaning load magnitude and will put more weight on the Backrest/Armrest_Junction XY coordinates.

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Please state some comments on what you *liked* about DA3.

Please state some comments on what you *disliked* about DA3.